Paths are Subspaces

a cut-triggered polymorphism of RSpace over path-keys, and its distributive law

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Abstract

The RSpace denotation of the rho calculus in the knotted universe factors into two independent pieces: a reflection structure on the *values* (the colour-swap s, with $@ = \kappa \circ s$ and $* = s \circ \kappa^{-1}$) and a store monoid on the *keys* ($\mathrm{Name} \rightharpoonup M_{\mathrm{fin}}(-)$ under key-wise $\uplus$). Anything done to the key object therefore leaves the reflection law untouched, so RSpace is polymorphic in its key type at no cost on the reflection side. We take the keys to be *paths*, $K = \mathrm{Name}^*$, the free monoid on names, whose store is a trie. The prefix order on paths supplies a notion of one channel lying in the *subspace* of another, and we read communication accordingly: a consume at path_1 and a produce at path_2 react when the paths are comparable, not merely equal. We insist throughout on a single discipline — *no implicit interaction*: the store never reacts with itself; a reaction is triggered only by a cut, an explicit $|$ co-locating a consume and a produce, exactly as in the source calculus. What comparability changes is not *when* a redex exists but *what* a given cut matches and *what residual* it leaves; and because the residual of a deep cut is again a single cut, one interaction can cause nested interactions. We give COMM in this generality as a pair of dual rules, exhibit the cascade as the reflection law $s \circ s = \mathrm{id}$ unsealing a located sub-RSpace, and write the dynamics as an explicit distributive law of the cut over the store monad, a bialgebra in the Turi–Plotkin sense, in which the unit axiom *is* the no-implicit-interaction principle. We then check the ledger: congruence by construction, the exact monoid homomorphism, OSLF adequacy and full abstraction, and the source paper's risk accounting all survive, with comparability the only new ingredient in the zip. Where the paths share prefixes — which, because names are quoted processes, is whenever channels share subterm structure — the trie is a radix-compressed, structurally shared, Merkle-addressable store, and we record where each compression pays off.

1 Introduction

The companion note [1] models the reflective higher-order (rho) calculus [3] inside the knotted universe of [2], with the quote @ reading as the colour-swap that seals a process into an atom of the opposite colour and the dereference $*$ as unsealing it. The dynamics there factor through an abstraction, *RSpace*: parallel composition is interpreted as keyed multiset union, so the denotation $[\![-]\!]$ is a monoid homomorphism and the parallel laws of structural congruence become literal equalities of tables, hashtables from channels to polarity-homogeneous bags of produces and consumes; communication is eager reaction, the zip of dual bags at a key, and RSpace is the store monad together with this zip, a bialgebra — the precise content of "strictly more than a monad".

The keys of that store are names, and in rho a name is a quoted process; the keys therefore have structure. This note studies one polymorphic interpretation of RSpace obtained by varying that structure. We make the key object a *path*: $K = \mathrm{Name}^*$, the free monoid on names, each key a route through a trie to a sub-RSpace. The interaction algebra of RSpace acts on entire RSpaces, so once keys carry a path order one may ask reaction to fire not only at equal keys but at *comparable* ones — a consume in the subspace addressed by a prefix of the produce's path, or *vice versa*.

One design decision governs the development.

No implicit interaction. The store is inert. It never reacts with itself, and there is no ambient broadcast in which a stored produce finds every comparable consume in the trie. A reaction is triggered only by a *cut*: an explicit parallel composition $|$ that co-locates a consume and a produce, exactly as $K =|$ is the single base constructor heading the left-hand side of COMM in the source calculus. What the path order buys is not new triggers but a richer *matching* at an existing trigger: $\mathsf{for}(y \leftarrow \mathrm{path}_1)\, P \mid \mathrm{path}_2\, !(Q)$ is a local cut even when path_1 and path_2 are not equal on the nose, and the comparability of the two paths decides what is delivered and what residual remains. Because the residual of a deep cut is again a single cut, an interaction can *cascade*: it can manufacture nested interactions, each of which is itself a cut, with nothing implicit in between.

This decision is what keeps the construction conservative. We show below that it leaves the locality of the cut — the reactive contexts are still $[\cdot] \mid R$ — and hence the congruence-by-construction argument, the exact homomorphism, OSLF adequacy, and the risk ledger of [1] all intact; the only genuinely new object is the comparability-aware zip, which we write out as a distributive law of the cut over the store monad. We assume throughout the modern presentation of [1]: $@P$ for quoting, $*x$ for dereference, $\mathsf{for}(y \leftarrow x)\, P$ for input, $x\,!(Q)$ for output, $\equiv$ for structural congruence, $\sim$ for context bisimilarity, $\sim_r$ for resource bisimilarity, and $[\![-]\!]\colon \mathrm{Proc} \to T(\mathrm{Proc})$ for the RSpace denotation into tables $T(R) = \mathrm{Name} \rightharpoonup \big(M_{\mathrm{fin}}(R) \uplus M_{\mathrm{fin}}([N]R)\big)$.

2 RSpace, parametric in the key object

2.1 The factoring lever

Write RSpace's table functor with its key object exposed as a parameter:

$$T_K(R) \;=\; K \;\rightharpoonup\; \big(M_{\mathrm{fin}}(R) \;\uplus\; M_{\mathrm{fin}}([N]R)\big),$$

a finite partial map from keys to polarity-homogeneous bags of payloads — a key holds either a multiset of output payloads (produces, data) or a multiset of input continuations (consumes, abstractions $(y)P$), never a mixture. The source paper is the instance $K = \mathrm{Name} \cong \mathrm{Proc}$, the keys quoted processes. The construction splits cleanly:

Remark 1 (The reflection/store factoring)**.** RSpace is two independent structures glued at the table. The first is a reflection on the *values*: the colour-swap $s\colon \mathcal{S} \cong \overline{\mathcal{S}}$, $s \circ s = \mathrm{id}$, with $@ = \kappa \circ s$ and $* = s \circ \kappa^{-1}$, validating $*@P \equiv P$ as the involutivity of s. The second is a store monoid on the *keys*: $K \rightharpoonup M_{\mathrm{fin}}(-)$ with multiplication $\uplus$, the free commutative monoid lifted key-wise. The reflection law concerns names, not the store monoid, and is undisturbed by any choice of K; the store laws concern keys, not values, and are undisturbed by any choice of value-reflection. Consequently every key-object polymorphism of RSpace validates Proposition (reflection) and clause (i) of the adequacy theorem of [1] *verbatim*, for free. The entire content of changing K lives in the store-and-zip factor.

2.2 What a key object must supply

The minimal interface to instantiate T_K as a store is decidable equality on K: enough to detect that a key carries both polarities, i.e. to recognise a redex. To do *path* reaction one asks for more

— the free monoid $(\mathrm{Name}^*, \cdot, \varepsilon)$ and its prefix order. To retain the congruence-by-construction of [1] one asks, finally, that the category of K-contexts possess the relative pushouts of Leifer–Milner [5, 6], adapted to GSLTs [4]; §6 shows the no-implicit-interaction discipline is exactly what makes this no harder for paths than for bare names.

3 Paths and the trie

3.1 The path key object

Fix $K = \mathrm{Name}^*$, finite sequences of names, with ε the empty path, $\cdot$ concatenation, $p \sqsubseteq q$ the prefix order ($q = p \cdot w$ for some suffix w), and $p \wedge q$ the longest common prefix, which exists for all p, q because the prefix order is a tree order. Write

$$p \asymp q \;:\Longleftrightarrow\; p \sqsubseteq q \;\text{ or }\; q \sqsubseteq p$$

for *comparability*: one path lies on the route to the other. (We reserve # for the freshness of [1].) The monoid acts on tables by descent: for $p \in K$ and a table u, the *graft* $p \cdot u$ is the table with $(p \cdot u)(p \cdot k) = u(k)$ and undefined off $p \cdot K$, and the *residual* u/p is the sub-table rooted at p, $(u/p)(k) = u(p \cdot k)$. A located output is sugar: $w \rhd Q$ is the process with

$$[\![w \rhd Q]\!] \;=\; \big[\, w \mapsto \{\!|\, [\![Q]\!] \,|\!\}^{+} \,\big], \qquad \varepsilon \rhd Q \equiv Q,$$

i.e. Q planted at sub-path w; since names are quoted processes, $@(w \rhd Q)$ is an honest name. Dually ε is a transparent rendezvous: an input or output on the empty path is on the node itself.

3.2 The store is a trie, and the trie compresses

Modulo $\equiv$ the parallel fragment is the free commutative monoid, and interpreting $|$ as key-wise $\uplus$ over the index Name^* keeps $[\![-]\!]$ a monoid homomorphism on the nose exactly as before. The store $\mathrm{Name}^* \rightharpoonup M_{\mathrm{fin}}(-)$ is, with prefix sharing, a trie:

$$\mathrm{Trie}(V) \;\cong\; \big(\mathrm{Name}^* \rightharpoonup V\big), \qquad \mathrm{Trie}(V) \;\cong\; V \times \big(\mathrm{Name} \rightharpoonup \mathrm{Trie}(V)\big),$$

the isomorphism on the left being precisely the statement that prefix sharing is sound: a trie is the representation of a path-indexed map in which common prefixes are stored once. The recursive presentation on the right makes every internal node itself a full RSpace — its own bag plus its children — which is what licenses "interaction acts on entire RSpaces" for nested stores.

Remark 2 (Four readings)**.** The path structure admits, in increasing order of how much of the monoid the reaction sees: *(0)* paths as opaque compound names — reaction at equal full paths, the trie a storage representation only, every theorem of [1] transferring verbatim; *(1)* the nested trie store with exact-key reaction — same behavioural and resource equivalences, but the recursion now exposed in the type and $\uplus$ realised as recursive trie-merge; *(2)* subspace reaction — a cut whose paths are comparable matches across the prefix order; and *(3)* meet reaction — a cut rendezvous at $p \wedge q$ even for incomparable siblings. Readings 0 and 1 are representational. Readings 2 and 3 change the matching, and are the subject of §4–§5. We develop the subspace reading in full and treat the meet reading as its symmetric extension.

4 Cut-triggered subspace reaction

4.1 COMM in the subspace, as a pair of dual rules

Communication is triggered by a cut and nothing else. When the two paths of a cut are comparable, the single COMM rule of the source calculus,

$$\frac{x \equiv_N z}{\mathsf{for}(y \leftarrow x)\, P \mid z\,!(Q) \longrightarrow P\{@Q/y\}}\ \textsc{Comm},$$

splits according to which side is the deeper. Let the input be on path_1 and the output on path_2.

Definition 1 (Subspace COMM)**.** There are two dual reactions, each a local cut.

(Out-deep) If $\mathrm{path}_1 \sqsubseteq \mathrm{path}_2$, write $\mathrm{path}_2 = \mathrm{path}_1 \cdot w$. The output lies in the subspace named by the input; the input receives the still-located sub-RSpace, sealed:

$$\mathsf{for}(y \leftarrow \mathrm{path}_1)\, P \mid \mathrm{path}_2\,!(Q) \ \longrightarrow\ P\{\, @(w \rhd Q) \,/\, y \,\}.$$

(In-deep) If $\mathrm{path}_2 \sqsubseteq \mathrm{path}_1$, write $\mathrm{path}_1 = \mathrm{path}_2 \cdot w$. The output arrives at the shallower channel and the input, waiting deeper, must descend w into the payload; the cut *unpacks* into a fresh, deeper cut:

$$\mathsf{for}(y \leftarrow \mathrm{path}_1)\, P \mid \mathrm{path}_2\,!(Q) \ \longrightarrow\ \mathsf{for}(y \leftarrow w)\, P \mid Q.$$

Both collapse to Comm at $w = \varepsilon$: (Out-deep) gives $P\{@Q/y\}$ since $\varepsilon \rhd Q \equiv Q$, and (In-deep) gives $\mathsf{for}(y \leftarrow \varepsilon)\, P \mid Q \longrightarrow P\{@Q/y\}$ by the transparency of the root rendezvous. The propagation rules Par, Struct are unchanged.

The right-hand side of (In-deep) is again a single parallel composition, a local cut, not a probe of the store. This is the precise sense of "an interaction can cause nested interactions": one cut produces a residual that is itself a redex, fired by the same rules, with nothing ambient in between. Whether it cascades depends on Q: if Q outputs on a channel comparable to w the new cut fires; otherwise it is a pending input, correctly stuck.

4.2 Reflection is the cascade engine

The seal of (Out-deep) is where reflection does double duty.

Proposition 1 (Cascade via $s \circ s = \mathrm{id}$)**.** *In* (Out-deep) *the receiver binds the name* $@(w \rhd Q) = \kappa\, s(w \rhd Q)$*, the located sub-RSpace sealed by the swap. Wherever the continuation* P *dereferences it,* $*$ *unseals it,*

$$*\,@(w \rhd Q) \ = \ s\,\kappa^{-1}\kappa\, s\,(w \rhd Q) \ = \ s\,s\,(w \rhd Q) \ = \ w \rhd Q,$$

splicing the located output back into the body of P*, where it may cut with* P*'s own consumes one level deeper. The colour-swap therefore fires twice per cascade step: once as the seal* $@ = \kappa \circ s$ *packaging the residual sub-RSpace into the delivered name, once under* $* = s \circ \kappa^{-1}$ *unsealing it for the next reaction.*

The cascade is thus not a new nonlocal mechanism but the reflection-through-reflection recursion the source calculus already relies on for recursion without a primitive replication operator (the server $D \equiv \mathsf{for}(y \leftarrow x)\, (*y \mid x\,!(y)) \mid x\,!(@(\cdots))$ of [1]), now also threading the trie.

5 The distributive law

We now present the dynamics of Definition 1 as a distributive law of the cut over the store monad, in the bialgebra format of Turi–Plotkin [7]. Two facts orient the construction: the store monad is untouched (still $\mathrm{Name}^* \rightharpoonup M_{\mathrm{fin}}(-)$ with $\uplus$), and the no-implicit-interaction principle is, formally, the *unit* axiom of the law.

5.1 The store monad and the pre-tables

Let (T, η, μ) be the store monad, $\mu = \uplus$, η the singleton placement, on the value functor of §2. During reaction a key may transiently carry both polarities, so we work on *pre-tables*

$$\overline{T}(R) \;=\; K \;\rightharpoonup\; W(R), \qquad W(R) \;=\; M_{\mathrm{fin}}(R) \times M_{\mathrm{fin}}([N]R),$$

a key holding a produce-bag B^+ and a consume-bag B^-; a table is the polarity-homogeneous subobject (one bag empty at every key). The cut installs both by key-wise union: $[\![P \mid Q]\!] = [\![P]\!] \uplus [\![Q]\!]$, with $\uplus$ pure (no firing) so that $[\![-]\!]$ stays a function and reaction is the separate dynamics.

5.2 The fire of one matched pair

A *consume item* is $(p, (y)P)$ and a *produce item* is (q, Q); they form a redex iff $p \asymp q$. Firing a matched pair returns the denotation of the corresponding rule's right-hand side, to be merged back by $\uplus$.

Definition 2 (Fire)**.** For $p \asymp q$ set w by $q = p \cdot w$ (when $p \sqsubseteq q$) or $p = q \cdot w$ (when $q \sqsubseteq p$), and define

$$\mathrm{fire}\big((p,(y)P),(q,Q)\big) \;=\; \begin{cases} [\![P\{\,@(w \triangleright Q)/y\,\}]\!] & p \sqsubseteq q \quad (\textsc{Out-deep}),\\ [\![\mathsf{for}(y \leftarrow w)\,P]\!] \;\uplus\; [\![Q]\!] & q \sqsubseteq p \quad (\textsc{In-deep}). \end{cases}$$

At $p = q$ both lines equal $[\![P\{@Q/y\}]\!]$, the source paper's COMM. The In-deep line is a fresh cut: $[\![\mathsf{for}(y \leftarrow w)\,P]\!] = [\,w \mapsto \{\!|(y)[\![P]\!]|\!\}^-]$ unioned with $[\![Q]\!]$, eager reaction continuing within.

5.3 The cut term and the law

Reaction enters only through juxtaposition. Given pre-tables s, t, the cut formed by composing them is the eager one-pass cross-reaction of all comparable opposite-polarity pairs with one side from each operand.

Definition 3 (Cut term)**.**

$$\chi(s,t) \;=\; \biguplus\Big\{\,\mathrm{fire}(c,d)\;:\;c \text{ a consume item of } s,\; d \text{ a produce item of } t,\; c \asymp d\Big\} \;\uplus\; (s \leftrightarrow t),$$

where $(s \leftrightarrow t)$ is the symmetric term (consumes of t against produces of s), and at each key the matched pairs are zipped along chosen list orderings, the surplus of the longer bag remaining — the located analogue of the zip of [1], its pairing nondeterminism the source calculus's choice of which send meets which receive, now ranging over comparable rather than equal keys.

Let $\rho\colon \overline{T} \to \overline{T}$ denote eager reaction to a (subspace-)polarity-homogeneous normal form: fire every comparable opposite-polarity pair, including those created by merged products, until none remains. The bialgebra content is two equations.

Definition 4 (Distributive law of the cut over the store)**.** Let Σ be the signature functor whose binary component is the cut $|$. A distributive law

$$\lambda : \Sigma T \implies T\Sigma$$

of Σ over the monad (T, η, μ) is a natural transformation satisfying

$$\textit{(D1, unit)} \quad \lambda \circ \Sigma\eta = \eta\Sigma, \qquad \textit{(D2, mult.)} \quad \lambda \circ \Sigma\mu = \mu\Sigma \circ T\lambda \circ \lambda T.$$

On the cut component λ is realised by $\uplus$-with-cross-reaction: on a pair of stores (s,t) it returns their key-wise union enriched by the cut term, $s \uplus t \uplus \chi(s,t)$, with the located residuals of Definition 2 reinstalled and reaction iterated to the normal form.

Proposition 2 (The unit axiom is no-implicit-interaction)**.** *$\chi(s, \varnothing) = \varnothing = \chi(\varnothing, t)$, hence λ on a cut against the empty store is the bare injection: $\rho(s \uplus \varnothing) = \rho(s)$. Equivalently* (D1)*: cutting against a unit store causes no reaction. A reaction requires a cut that actually co-locates opposite polarities; the store never reacts with itself.*

Proof sketch. $\chi(s,t)$ ranges over pairs (c, d) with c from one operand and d from the other; with one operand empty there are no such pairs, so the cut term is empty and λ reduces to $\uplus$, whose unit is $\varnothing$. This is precisely the statement that $\Sigma\eta$ followed by λ equals $\eta\Sigma$: composing a process with the inert unit and then "reacting" is the same as not reacting. □

Proposition 3 (Multiplication, up to resource bisimilarity)**.** *For pre-tables s, t,*

$$\rho(s \uplus t) \ \sim_r \ \rho\big(\rho(s) \uplus \rho(t) \uplus \chi(s,t)\big).$$

The right-hand side is the (D2) *pattern: react the operands, merge, and add exactly the cross-cuts the union created; the cross-cuts are the only reactions not already internal to s or t, and they are exactly the comparable opposite-polarity pairs straddling the cut. Iterating ρ closes the cascade.*

Remark 3 (What kind of object this is)**.** The law does not change the monad: $\uplus$ is verbatim the store multiplication, and Proposition 13 of [1] (the exact homomorphism, the parallel laws as literal table equalities) holds unchanged because $\uplus$ never fires. RSpace remains "store monad + zip"; the zip is merely comparability-aware. The reflection law is likewise untouched, by Remark 1. The cascade is the inductive closure of λ over its own In-deep residuals; it terminates whenever the source process is finite, because each fire strictly consumes a matched pair, and the only route to nontermination is reflection looping — the single ω of §6, not a new infinity.

5.4 The meet variant

The pure-meet reading fires at $p \wedge q$ even for incomparable siblings, $p = m \cdot u$ and $q = m \cdot v$ with u, v both nonempty, $m = p \wedge q$. It remains cut-triggered. It carries one genuine degree of freedom the subspace reading does not: with neither path containing the other one must fix a convention for both residual suffixes, the natural symmetric move delivering each side's residual into the other and relocating the continuation under m. This is where the richest cascades live, and where one owes an explicit relocation operator; the subspace reading keeps a clean container/contained relationship and is the conservative one to develop first.

6 What transfers

We record, against the apparatus of [1], what the subspace dynamics preserves. The thread throughout is locality: because reaction is cut-triggered, the reactive contexts do not change.

Congruence, by construction. The reactive contexts are still $[\cdot] \mid R$ — a cut always lies at a parallel composition, including the manufactured right-hand cut of (In-deep) — so the category of contexts and its idem-pushouts never see the matching predicate. The relative-pushout hypothesis (Leifer–Milner [5, 6], GSLT minimal-context [4]) is no harder than for bare names, and context bisimilarity is a congruence by construction: $P \sim Q \Rightarrow F[P] \sim F[Q]$. The action-labelled closure-under-substitution question does not arise.

The label set, three views. The labels are still the parallel cuts ∂T_K; the OSLF-generated Hennessy–Milner modalities $\langle F\rangle\varphi$ are still indexed by the families $[\cdot] \mid \mathrm{path}_2\,!(Q)$, now with path_2 ranging over the comparables of the input path — a larger, but still context-shaped, family. Firing, Witnessing and Placing remain three readings of one label set, and OSLF adequacy gives $P \sim Q \Leftrightarrow \forall\varphi.(P \models \varphi \Leftrightarrow Q \models \varphi)$.

The homomorphism and the bialgebra shape. $\uplus$ is untouched, so the exact monoid homomorphism survives and the parallel laws are literal table equalities; reaction is a richer zip over the same store monad, the "store monad + zip" bialgebra of [1] with comparability-matching and located residuals the only changes. The polarity-homogeneity invariant of reduced tables generalises to its subspace form — no two *comparable* keys carry opposite polarities — and no-implicit-interaction is exactly the claim that eager reaction maintains this invariant: every opposite-polarity comparable pair was introduced by a $|$, so firing it is sound and traceable to a cut.

Full abstraction and the risk ledger. The factoring of [1] onto νB persists: resource bisimilarity (counting located copies) refines context bisimilarity, with the projection π forgetting multiplicity and keys. Full abstraction $[\![P]\!]_B = [\![Q]\!]_B \Leftrightarrow P \sim Q$ holds for a congruence with no closure step. And the risk ledger is undisturbed: each cut is finite, a cascade from finite terms is finite, and the only source of nontermination is reflection looping, the same single ω — black infinity, an ω-fold multiplicity — the source paper already charges. The path machinery spends no new infinity; it routes the existing reflective recursion through the trie.

7 Where the trie compresses

Four mechanisms, in increasing order of how much they touch the semantics rather than the representation.

Radix (Patricia) collapse [9]. Stores are sparse, so the trie is mostly chains of single-child nodes; collapsing each chain into one edge labelled by a path segment takes the depth from one node per name to one node per branch point. A storage win, available in every reading.

Structural sub-trie sharing (hash-consing, copy-on-write). Eager normal forms are subspace-polarity-homogeneous: a reduced sub-trie is inert, with no pending redex, hence shareable by reference and duplicated lazily only when one side reacts. This is where the recursive trie-merge of Reading 1 pays, and it is the operational face of the resource layer: copy-on-write is "do not materialise a multiplicity until it is observed."

Merkle addressing. Tag each node with the hash of its children and sub-RSpaces become content-addressed. Resource bisimilarity (a key-by-key, multiplicity-matched comparison of tables) then short-circuits equal subtrees in $O(1)$ — equal root hash means $\sim_r$-equal sub-store, no descent — and content addressing is native, not bolted on, because Name = @Proc: the key already *is* a hash of a quoted process.

The path is the structure of the name. Since names are quoted processes, the natural path is not an arbitrary string but the spine of the quoted process itself — its parallel decomposition, its descent through the term. Then content-addressed channels that share *subterms* share *trie structure*, automatically, by the structure of the quote; and large quoted-process names, the case where opaque keys waste the most, are exactly where the de-duplication is greatest. This is the denotational reading of what the on-chain RSpace history store already computes when it persists to a Merkle radix trie keyed by hashes of channels: Readings 1–3 are the semantics that implementation has been evaluating all along.

8 Obligations

In the manner of [1], §8, the points a fully rigorous development would discharge.

Obligation 1 (The distributive law)**.** Verify *(D1)* and *(D2)* for λ of Definition 4 in the Turi–Plotkin format [7], with $[\![-]\!]$ the induced bialgebra morphism. *(D1)* is Proposition 2; *(D2)* is the coherence of cut formation across nested unions, of which Proposition 3 states the $\sim_r$-form. Confirm that the located residuals of Definition 2 make λ natural in R.

Obligation 2 (Eager confluence up to $\sim_r$)**.** A shallow consume may be comparable to several deeper produces (fan-out) and a deep input unpacks into a payload (descent), so eager reaction must be shown confluent up to resource bisimilarity with the new residuals: all maximal reaction sequences reach $\sim_r$-equal normal forms, independent of the pairing nondeterminism of Definition 3 and the fan-out order.

Obligation 3 (Substitution and relocation)**.** Prove $[\![P\{@(w \rhd M)/y\}]\!] = [\![P]\!]\{@(w \rhd [\![M]\!])/y\}$, the located generalisation of the substitution lemma of [1], carried by the abstraction functor's evaluation $[N]R \times \mathrm{Name} \to R$, with the seal $\kappa \circ s$ applied to the message. For the meet variant, establish the relocation operator that delivers each incomparable residual into the other and pin down whether it is primitive or derivable.

Obligation 4 (Relative pushouts for path-contexts)**.** Discharge the locality claim of §6: the category of rho contexts over path-keys, with plugging as composition, possesses relative pushouts, so the idem-pushout labels are well defined and the Leifer–Milner congruence theorem applies. The cut lying always at a parallel composition (including the In-deep residual) and names being quoted structure rather than ν-bound are the features that should make the relative pushouts exist and be computable.

Obligation 5 (Subspace-homogeneous normal forms)**.** Show eager reaction maintains the subspace-polarity-homogeneity invariant — no two comparable keys carry opposite polarities in a normal form — and that this is equivalent to no-implicit-interaction: every comparable opposite-polarity pair is traceable to a cut.

Obligation 6 (The trie store in the knotted universe)**.** Encode the trie inside the knotted universe so that recursion is accommodated and the swap s and knots $\kappa, \bar{\kappa}$ respect the path structure and the count arithmetic, multiplicity-ω being exactly the black infinity of §6; confirm T_{Name^*} preserves weak pullbacks (the index-set change to Name^* leaves the refinement-monoid argument of [8] intact, the $\mathbb{N} \cup \{\omega\}$ completion to be re-checked).

9 Conclusion

Reading RSpace's keys as paths makes the store a trie and the prefix order a notion of one channel lying in another's subspace. Insisting that reaction stay cut-triggered — the store never reacts with itself, an interaction requires a $|$, and an interaction may manufacture nested interactions — turns the subspace reading into a conservative extension: COMM splits into the dual OUT-DEEP/IN-DEEP rules, the cascade is the reflection law $s \circ s = \mathrm{id}$ unsealing a located sub-RSpace, and the dynamics is a distributive law of the cut over the unchanged store monad whose unit axiom is precisely the no-implicit-interaction principle. The cost is paid on the confluence side, where it is the same genus of rewrite obligation the eager semantics already carries; the congruence side, the exact homomorphism, OSLF adequacy and full abstraction, and the risk ledger are all preserved. And because names are quoted processes, the path can be the structure of the name, so the trie is not only a storage compression but the denotational content of the Merkle radix store the implementation already runs.

Future work. The meet variant and its relocation operator; the lazy reaction strategy, treating a comparable opposite-polarity pair as a standing redex rather than reducing eagerly; and the join-pattern axis — paths grade the store by the free monoid Name^* under concatenation, whereas join patterns grade it by tupling Name^n, and the two gradings compose.

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